Scientific Protocols: the Making and Breaking of Scientific Norms | Area: General Philosophy of Science

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38 Scientific Protocols: the Making and Breaking of Scientific Norms

Short Description

Scientific protocols are a ubiquitous element of knowledge production and evidence-based policy. From laboratory instructions to disease diagnostics and community engagement, scientists and practitioners regularly operate upon standardized, agreed upon, explicit procedures. Our presentations are motivated by three general questions: How do protocols concretize abstract norms into specific instructions? What are virtues and limitations of protocols as norms ambivalent between stability and variability? What kind of entities are protocols? Guided by these questions, we present on measurement and model-preparation protocols in molecular biology (Bollhagen); implementation and stabilization of single-cell omics protocols and their epistemic implications (Ferree); protocol variation as a means to make different systems comparable in cell culture research (Yoshida); how standardization of diagnostic protocols leads to exclusion of patients' lived experience (Alcalay); and protocols' role in facilitating community engagement in interdisciplinary conversations (Kenis and Bocchi). This symposium will call attention to the epistemic and normative attributes of scientific protocols in connection with timely issues in philosophy of science, such as reproducibility and epistemic justice, demonstrate the value of focusing on protocols for conducting philosophical research based firmly on concrete practices, and appeal to PSA attendees broadly by providing analytical perspectives applicable across domains and contexts of science.

Participants

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Individual Paper Abstracts

Developing Protocols for Explanatory Modeling and Measuring: Insights from the Practice of *In Vitro* Reconstitution

Andrew Bollhagen

A protocol is a detailed set of steps designed to standardize the execution of a procedure to achieve some particular end. For example, a recipe, if followed carefully, will yield a tasty dish. *In vitro reconstitution* is an experimental practice that "seeks to recapitulate biological events outside their natural context using a reduced number of parts (Schlissel *et al.* 2020). In general, protocols for constructing *in vitro* systems involve isolating elements from a living system and attempting to produce a phenomenon of interest "in glass" (hence, *in* vitro). As the cases that I discuss show, however, the ontological principles that govern the epistemic activity of

developing these protocols differ depending upon whether the *in vitro reconstituted system* (IVRS) serves as an *explanatory model* or as an *instrument of measure* (Chang 2009).

My talk offers a comparative analysis of the development of two such protocols: 1) a protocol for preparing an IVRS as an *model* to *explain* muscle contraction and 2) a protocol for preparing an IVRS to *measure* the characteristic rate at which proteins involved in muscle contraction (and many other cellular phenomena)—myosin and actin—move relative to one another. The two cases are closely related. Both involve using the same kinds of proteins (myosin and actin) to produce observable motion "in a petri dish." Both cases illustrate that IVRSs can be produced following either a "bottom up" or "top down" protocol. Further, whether the IVRS is to be used as a model or an instrument of measure, it is important for them to be "biochemically well-defined," meaning that researchers know the identities of all the elements present in the preparation such that they can be confident that *just these elements* are responsible for the system's behavior. Finally, in both cases, the development of the protocol involved a kind of "bootstrapping" by which iterations of the protocol were formulated, criticized, and improved upon. However, I point out these similarities only to help to throw into relief the differences. As I show, researchers appeal to quite different evaluative criteria in criticizing and, thereby, driving the bootstrapping process depending on whether the protocol under development aims at producing an IVRS to serve as an adequate *explanatory model* or *instrument of measure*. Following Hasok Chang (2022), I identify the "ontological principle" that governs the "epistemic activity" of developing IVRS modeling protocols as Leibniz's Law (identity of indiscernbles) while the ontological principle that governs the epistemic activity of developing measurement protocols is the principle of single-value.

Protocol Variation and the Methodological Principle of Invariance

Yoshinari Yoshida

Standardization and variation are both prevalent features of scientific protocols. On the one hand, a protocol instructs us to follow a very specific procedure. Protocols operationalize the methodological principle of invariance—the idea that one should identify a parameter or variable that is kept fixed across systems, groups, or cases to make them comparable (Leonelli and Tempini 2021). On the other hand, using a protocol in a particular context often requires adjustments and modifications of certain aspects of it (Berg 1998; Boem 2020; Smolka 2023). In this context, variability of protocols is typically understood as a practical issue that occurs due to local contexts *despite* the ideal of invariance.

In this presentation, I argue that variation of protocols is sometimes *warranted* by the principle of invariance. To illustrate this, I focus on cross-species comparisons of *in vitro* models as a case study. In this emerging area, scientists compare cell cultures of different species to acquire insights into similarities and differences between development and physiology of the donor species. When conducting such a cross-species comparison on a dish, scientists seek to apply an identical protocol to culture cells of different species to ensure that any differences found between them can be attributed not to variations of experimental settings, but to the species difference. But in some cases, scientists choose to vary protocols between *in vitro* models, reflecting different needs and requirements of cells of different species. I argue that this experimental approach can be understood as aiming at *invariance in states or stages* of *in vitro* models, instead of *invariance in external conditions* surrounding cells. This consideration suggests that there are two kinds of protocol practices to pursue invariance in comparative studies: (i) adopting identical protocols to compare systems under as similar conditions as possible, and (ii) varying protocols to provide each system a suitable condition so that they reach equivalent states. I discuss how scientists choose between the two approaches. While the latter approach could enable a more physiologically relevant comparison, it requires scientists to interpret and justify the meaning of invariant states or stages in each case. The former approach provides a conceptually much more straightforward instruction, which is why it is typically a default option. Importantly, the above methodological issue occurs across fields of science; I illustrate this by discussing some examples from comparative cognition research.

Creeping Doubt of Chronicity

Rena Alcalay

Clinical guidelines in endocrinology function as a specific type of protocol that has largely escaped philosophical scrutiny. Yet analyzing them through this lens reveals how such protocols can systematically generate epistemic disadvantage and injustice. This paper argues that endocrine diagnostic protocols, which prioritize quantification and algorithmic predictions over lived experience, constitute a form of norm-governed practice that excludes certain kinds of experience from legitimacy, which has concrete epistemic consequences for patients.

The central case concerns baseline measurement in the diagnosis of thyroid disease. Although thyroid-stimulating hormone (TSH), thyroxine (T4), and triiodothyronine (T3) are quantitatively measured within ostensibly fixed reference ranges, significant diagnostic uncertainties persist. Baseline ranges lack convergence (Soga and Gaston, 2018; Linares, 2022), normative frameworks shape interpretation (Alexandrova and Fabian 2022; Muller 2018; Sepkoski 2020), and crucially, symptoms can easily fail to align with clinical baselines. This misalignment reflects how survivor bias in predictive systems symmetrically excludes symptoms that resist population-level categorization.

I argue that survivor bias in endocrinology constitutes epistemic disadvantage (Goldstein 2022): a structurally rooted harm distinct from epistemic injustice, in which institutional systems treat quantifiable markers as the sole arbiters of credibility, thereby excluding patient testimony and bodily doubt—the legitimate uncertainty patients experience when their lived illness diverges from biomedical markers (cf. Carel 2013, 2025). When biases are embedded in the level of clinical measurement and amplified by institutional structures that prioritize standardization, they create a feedback loop in which doubt itself becomes pathologized rather than recognized as epistemically legitimate.

By examining clinical guidelines as protocols, this paper contributes to the symposium's broader investigation into how norms stabilize or shift across scientific settings, and what epistemic and social functions protocols serve. It raises an urgent question for endocrinology and metabolic practice: how might protocols be redesigned to acknowledge patient experience not as subjective noise to be filtered out, but as especially necessary for responsible clinical practice in the treatment of autoimmune conditions.

The Non-Epistemic Dimension of Scientific Protocols

Daan Kenis & Federica Bocchi

The few available philosophical accounts of scientific protocols have emphasized how standardized procedures guide laboratory work and data collection, likening protocols to recipes for robust science (Boem 2016, Bocchi 2025). Yet, not all scientific protocols are confined to the lab or field. Some are designed to structure human interactions, facilitate collaboration in scientific settings, and foster community, representing bona fide *knowledge-promoting* scientific practices (Currie 2019). These protocols extend beyond epistemic outcomes and instead are developed to shape the research ecosystem itself: they create conditions that *indirectly* contribute to scientific advancements.

We examine the conditions under which *community-engagement protocols* are developed and implemented, drawing on two case studies from distinct domains each illustrating how protocols can foster inter- and trans-disciplinary collaborations.

The first case examines a consensus procedure employed in a transdisciplinary biomedical autism research project: the PARADIGM Patient Engagement toolbox. Here, neurodivergent individuals and biomedical researchers collaborate to address challenges rooted in deficit-focused autism research. The consensus protocol harmonizes retrospective cohort data using classifications that better reflect the needs of neurodivergent individuals, while also aiming to improve clinical diagnostics and interventions (Black et al. 2024).

The second case involves CoNavigator (Lindvig et al. 2016, 2025), a toolkit designed to facilitate interdisciplinary collaboration in academic settings. In multi-hour sessions, participants from diverse academic backgrounds articulate key disciplinary concepts and collaboratively develop research clusters that integrate—without assimilating—multiple perspectives. By making explicit the assumptions and knowledge each participant brings, CoNavigator bridges disciplinary divides and fosters creative, mutual understanding, necessary for addressing so-called "wicked problems."

We emphasize that community building is a central goal of these protocols, and defend their status as genuinely scientific—rather than merely social—norms. Achieving these aims requires that protocols embed social values: they enable individuals to participate actively in deliberation, demonstrate trustworthiness, and practice (disciplinary and cultural) epistemic humility. We draw three corollaries regarding their communitarian, norm-building character:

1. Scientific protocols can help prevent contributory injustice (Dotson 2012) and scientific imperialism in inter- and transdisciplinary collaborations at the institutional level (Lam & Rousselot 2024). By including mechanisms to break down hierarchies, such protocols ensure that dominant disciplinary standards no longer set the default.
2. Protocols may actively promote epistemic repair by facilitating value alignment among stakeholders. By enabling participants to play a substantive role in shaping research, protocols may directly contribute to scientific institutions' epistemic trustworthiness. The structure of the protocol itself, rather than its implementation or outcomes alone, can thus foster epistemic justice.
3. Collaboration-enhancing protocols often treat effectiveness of scientific collaboration as a proxy for a stronger epistemic outcome. However, even if no tangible epistemic good results, the protocols discussed may inform a successful intervention by exposing structural barriers to epistemic justice.

By focusing on protocols' non-epistemic aims, we highlight an underexplored dimension in the philosophy of science and contribute to ongoing discussions about the roles of norms, values, and epistemic injustice within science.

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Categories

General Philosophy of Science